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#Name - vidya Rajan Dharne.
#Roll No - CS08
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# Aim - To familiarize students with Google Colab and Python libraries (NumPy, Pandas, Matplotlib
#and Seaborn) for loading, inspecting, cleaning, manipulating and visualizing the Titanic
#dataset.
```

```
# Task 1: Import Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from google.colab import files
import pandas as pd

# Upload the CSV file
uploaded = files.upload()

# Read the CSV file
df = pd.read_csv("train.csv")

# Display the first five rows
df.head()
```

Choose files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving train.csv to train.csv

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S

```
#Task 3: Display first 10 rows using df.head(10).
df.head(10)
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
5	6	0	3	Moran, Mr. James	male	NaN	0	0	330877	8.4583	NaN	Q
6	7	0	1	McCarthy, Mr. Timothy J	male	54.0	0	0	17463	51.8625	E46	S
7	8	0	3	Palsson, Master. Gosta Leonard	male	2.0	3	1	349909	21.0750	NaN	S

```
#Task 4: Display last 5 rows using df.tail(5).
df.tail(5)
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.75	NaN	Q

```
#Task 5: Check dataset dimensions using df.shape (Expected: 891 x 12).
df.shape
```

(891, 12)

```
#Task 6: Display dataset information using df.info().
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   PassengerId      891 non-null    int64
1   Survived         891 non-null    int64
2   Pclass           891 non-null    int64
3   Name             891 non-null    object
4   Sex              891 non-null    object
5   Age              714 non-null    float64
6   SibSp            891 non-null    int64
7   Parch            891 non-null    int64
8   Ticket           891 non-null    object
9   Fare             891 non-null    float64
10  Cabin            204 non-null    object
11  Embarked         889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
#Task 7:
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
#Task 8: Identify missing values using df.isnull().sum().
df.isnull().sum()
```

```
0
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age           177
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin         687
Embarked       2
```

dtype: int64

```
#Task 9: Fill missing Age values with the median.
print(df["Age"].isnull().sum())
```

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```
#Task 10: Filter surviving passengers (Survived == 1).
import pandas as pd

# Load dataset
df = pd.read_csv("train.csv")

# Filter surviving passengers
survived_passengers = df[df["Survived"] == 1]

# Display surviving passengers
survived_passengers
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
8	9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27.0	0	2	347742	11.1333	NaN	S
9	10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14.0	1	0	237736	30.0708	NaN	C
...	...	...	...	...	...	...	...	...	...	...	...	...
875	876	1	3	Najib, Miss. Adele Kiamie "Jane"	female	15.0	0	0	2667	7.2250	NaN	C
879	880	1	1	Potter, Mrs. Thomas Jr (Lily Alexenia Wilson)	female	56.0	0	1	11767	83.1583	C50	C

```
#Task 11: Convert Fare column to a NumPy array.
import pandas as pd
import numpy as np

# Load dataset
df = pd.read_csv("train.csv")

# Convert Fare column to NumPy array
fare_array = df["Fare"].to_numpy()
```

```
# Display array
fare_array
```

```
array([ 7.25 , 71.2833, 7.925 , 53.1 , 8.05 , 8.4583,
 51.8625, 21.075 , 11.1333, 30.0708, 16.7 , 26.55 ,
 8.05 , 31.275 , 7.8542, 16. , 29.125 , 13. ,
 18. , 7.225 , 26. , 13. , 8.0292, 35.5 ,
 21.075 , 31.3875, 7.225 , 263. , 7.8792, 7.8958,
 27.7208, 146.5208, 7.75 , 10.5 , 82.1708, 52. ,
 7.2292, 8.05 , 18. , 11.2417, 9.475 , 21. ,
 7.8958, 41.5792, 7.8792, 8.05 , 15.5 , 7.75 ,
 21.6792, 17.8 , 39.6875, 7.8 , 76.7292, 26. ,
 61.9792, 35.5 , 10.5 , 7.2292, 27.75 , 46.9 ,
 7.2292, 80. , 83.475 , 27.9 , 27.7208, 15.2458,
 10.5 , 8.1583, 7.925 , 8.6625, 10.5 , 46.9 ,
 73.5 , 14.4542, 56.4958, 7.65 , 7.8958, 8.05 ,
 29. , 12.475 , 9. , 9.5 , 7.7875, 47.1 ,
 10.5 , 15.85 , 34.375 , 8.05 , 263. , 8.05 ,
 8.05 , 7.8542, 61.175 , 20.575 , 7.25 , 8.05 ,
 34.6542, 63.3583, 23. , 26. , 7.8958, 7.8958,
 77.2875, 8.6542, 7.925 , 7.8958, 7.65 , 7.775 ,
 7.8958, 24.15 , 52. , 14.4542, 8.05 , 9.825 ,
 14.4583, 7.925 , 7.75 , 21. , 247.5208, 31.275 ,
 73.5 , 8.05 , 30.0708, 13. , 77.2875, 11.2417,
 7.75 , 7.1417, 22.3583, 6.975 , 7.8958, 7.05 ,
 14.5 , 26. , 13. , 15.0458, 26.2833, 53.1 ,
 9.2167, 79.2 , 15.2458, 7.75 , 15.85 , 6.75 ,
 11.5 , 36.75 , 7.7958, 34.375 , 26. , 13. ,
 12.525 , 66.6 , 8.05 , 14.5 , 7.3125, 61.3792,
 7.7333, 8.05 , 8.6625, 69.55 , 16.1 , 15.75 ,
 7.775 , 8.6625, 39.6875, 20.525 , 55. , 27.9 ,
 25.925 , 56.4958, 33.5 , 29.125 , 11.1333, 7.925 ,
 30.6958, 7.8542, 25.4667, 28.7125, 13. , 0. ,
 69.55 , 15.05 , 31.3875, 39. , 22.025 , 50. ,
 15.5 , 26.55 , 15.5 , 7.8958, 13. , 13. ,
 7.8542, 26. , 27.7208, 146.5208, 7.75 , 8.4042,
 7.75 , 13. , 9.5 , 69.55 , 6.4958, 7.225 ,
 8.05 , 10.4625, 15.85 , 18.7875, 7.75 , 31. ,
 7.05 , 21. , 7.25 , 13. , 7.75 , 113.275 ,
 7.925 , 27. , 76.2917, 10.5 , 8.05 , 13. ,
 8.05 , 7.8958, 90. , 9.35 , 10.5 , 7.25 ,
 13. , 25.4667, 83.475 , 7.775 , 13.5 , 31.3875,
 10.5 , 7.55 , 26. , 26.25 , 10.5 , 12.275 ,
 14.4542, 15.5 , 10.5 , 7.125 , 7.225 , 90. ,
 7.775 , 14.5 , 52.5542, 26. , 7.25 , 10.4625,
 26.55 , 16.1 , 20.2125, 15.2458, 79.2 , 86.5 ,
 512.3292, 26. , 7.75 , 31.3875, 79.65 , 0. ,
 7.75 , 10.5 , 39.6875, 7.775 , 153.4625, 135.6333,
 31. , 0. , 19.5 , 29.7 , 7.75 , 77.9583,
 7.75 , 0. , 29.125 , 20.25 , 7.75 , 7.8542,
 9.5 , 8.05 , 26. , 8.6625, 9.5 , 7.8958,
 13. , 7.75 , 78.85 , 91.0792, 12.875 , 8.85 ,
 7.8958, 27.7208, 7.2292, 151.55 , 30.5 , 247.5208,
 7.75 , 23.25 , 0. , 12.35 , 8.05 , 151.55 ,
 110.8833, 108.9 , 24. , 56.9292, 83.1583, 262.375 ,
 26. , 7.8958, 26.25 , 7.8542, 26. , 14. ,
 164.8667, 134.5 , 7.25 , 7.8958, 12.35 , 29. ,
 69.55 , 135.6333, 6.2375, 13. , 20.525 , 57.9792,
 23.25 , 28.5 , 153.4625, 18. , 133.65 , 7.8958,
 66.6 , 134.5 , 8.05 , 35.5 , 26. , 263. ,
 13. , 13. , 13. , 13. , 13. , 16.1 ,
```

```
#Task 12: Calculate maximum and average Fare using np.max() and np.mean().
import pandas as pd
import numpy as np
```

```
# Load dataset
df = pd.read_csv("train.csv")
```

```
# Convert Fare column to NumPy array
fare_array = df["Fare"].to_numpy()
```

```
# Calculate maximum and average Fare
print("Maximum Fare:", np.max(fare_array))
print("Average Fare:", np.mean(fare_array))
```

```
Maximum Fare: 512.3292
Average Fare: 32.204207968574636
```

```
#Task 13: Plot histogram of Age.
import pandas as pd
import matplotlib.pyplot as plt
```

```
import seaborn as sns

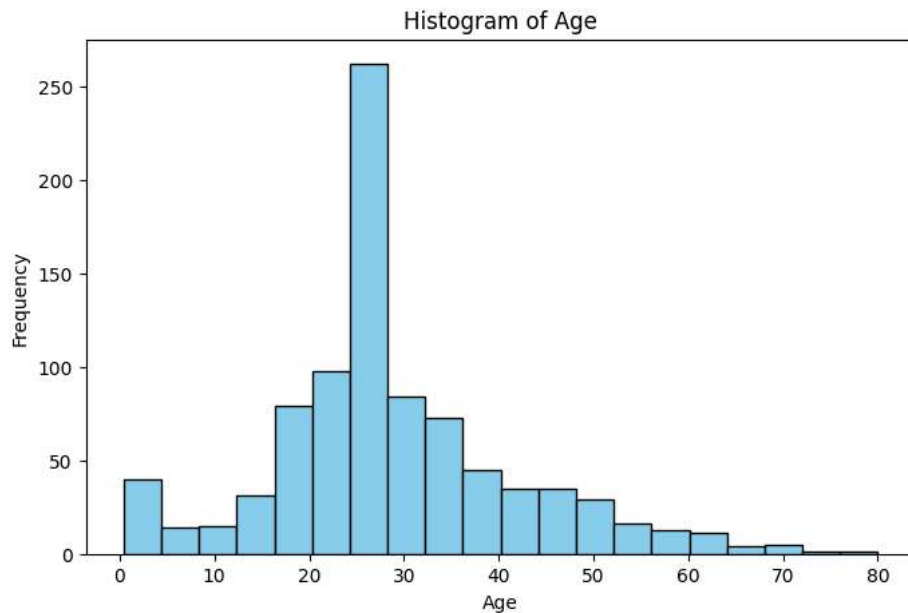
# Load dataset
df = pd.read_csv("train.csv")

# Fill missing Age values with median
df["Age"] = df["Age"].fillna(df["Age"].median())

# Plot histogram
plt.figure(figsize=(8,5))
plt.hist(df["Age"], bins=20, color="skyblue", edgecolor="black")

plt.xlabel("Age")
plt.ylabel("Frequency")
plt.title("Histogram of Age")

plt.show()
```



```
#Task 14: Plot bar chart of Survival Count.
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("train.csv")

# Count survival values
survival_count = df["Survived"].value_counts()

# Plot bar chart
plt.figure(figsize=(6,4))

sns.barplot(x=survival_count.index, y=survival_count.values, palette="viridis")

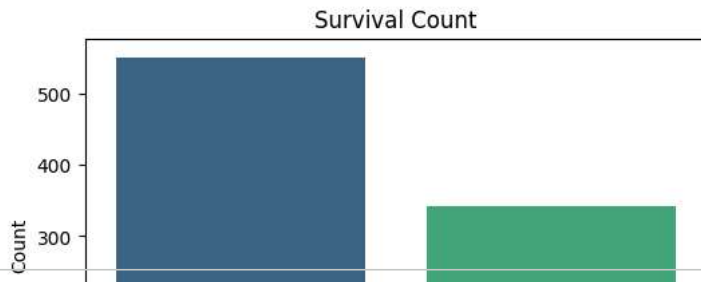
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Count")
plt.title("Survival Count")

plt.show()
```

/tmp/ipykernel_4983/2405894766.py:15: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

```
sns.barplot(x=survival_count.index, y=survival_count.values, palette="viridis")
```



#Task 15: Plot scatter graph of Age vs Fare.

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
# Load dataset
```

```
df = pd.read_csv("train.csv")
```

```
# Fill missing Age values with median
```

```
df["Age"] = df["Age"].fillna(df["Age"].median())
```

```
# Create scatter plot
```

```
plt.figure(figsize=(8,5))
```

```
plt.scatter(df["Age"], df["Fare"], color="blue", alpha=0.5)
```

```
plt.xlabel("Age")
```

```
plt.ylabel("Fare")
```

```
plt.title("Age vs Fare Scatter Plot")
```

```
plt.show()
```

